

The Evolution of Shot Selection in the NBA: Shot Zones Over Time By Position

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Background, Motivation, and Questions

The selection of this project was motivated by the widely observed shift toward three-point (3PT) shooting in the National Basketball Association (NBA). Previous studies have explored shot patterns following the implementation of the three-point line. One study examines the year-to-year changes, finding that differences in shot patterns happened due to modifications in the 3PT line distance (Freitas, 2020). Previous studies have also shown that 3PT attempts have significantly increased while 2PT attempts have decreased (Zajac et al., 2023). Some studies focus on the decrease in mid-range attempts and why that is happening (Kilcoyne, 2020). While we are similarly motivated by observations of increases in three-point shooting, we focus our efforts on this century and on shot selection changes by player position and shooting efficiency.

The following work explores how **shot selection in the NBA has evolved over time** and how potential shot trends may differ for guards, forwards, and centers. This project uses detailed shot-location data combined with player metadata to examine how the distribution of shots taken in three designated zones of “the paint” (i.e. near the basket), “mid-range”, and “three-point” areas has changed from the turn of the century to the present. This motivation led us to refine our efforts in order to address three questions:

1. How has the distribution of shots taken in the paint, mid-range, and three-point areas changed from the early 2000s to the present?
2. How do these shot selection trends differ for guards, forwards, and centers, and how have positional roles evolved over time?
3. If there are changes in shot selection found, are they accompanied by corresponding changes in shot efficiency across different zones and positions?

NBA Shot Zone Selection Analysis - Data Sources

Primary Dataset

NBA Stats API

Description: We use the NBA shot charts from NBA Stats API. This contains shot-level data for NBA games, where each recorded shot corresponds to an individual shot attempt.

Size: Approximately five million recorded shots across 24 seasons were used. We store the shot records we use in a dataframe that is 336 MB.

Variables: Season, Player Name, Shot location (Loc_X and Loc_Y), Shot Type, Shot Made Flag, Shot Zone Basic

Format: JSON responses to tabular form

Access: API

Location: NBA Stats API (via the nba_api Python package)

Secondary Dataset

Basketball Reference Player Metadata

Description: Provides basketball statistics and history, of which we use the NBA player metadata and stats to contextualize shot data and to group players into position categories.

Size: Thousands of player-season records were used from this source. We store the player-season records in a dataframe that is 350 KB.

Variables: Season, Player Name, Position, Team Name

Format: HTML tables scraped and converted to CSV

Access: Web Scraping

Location: Basketball Reference
(<https://www.basketball-reference.com/>)

Data Analysis Process Goals & Manipulation

NBA Stats API

Goal: Create a season-level NBA shot dataset using seasons from 2000-01 to 2023-24 with shot location, player and season data, and consolidate shot zones to analyze those we are interested in.

1. For every season between 2000-01 to 2023-24 we pulled every shot made by every player across the regular season resulting in one dataframe, shots_df.
2. We selected variables related to shot location, shot outcome, player and team identity, and season. We next dropped shot attempts with missing coordinates and chose to exclude backcourt shots.
3. Next we did feature engineering to consolidate the shots zones from six original categories [Restricted Area, In The Paint (Non-RA), Mid-Range, Above the Break 3, Left Corner 3, and Right Corner 3] into three meaningful categories of Paint, Mid-Range, and 3PT shots by mapping, and then removed the original column defining the original six zones.

Basketball Reference

Goal: Create a player-season dataset to collect player metadata, particularly position.

1. A list of seasons and end years was used to access Basketball Reference statistics tables, and pandas.read_html was used to extract HTML tables.
2. Player name, position, team name, and season variables were kept and each season was added to a player-season dataset.
3. Next we cleaned players who appeared on multiple teams within a season. For these players, we prioritize "TOT" or total, to be used in this dataframe instead of the player's data for one team or another, so each player-season combination will only appear once.
4. We consolidated positions into three groups: Guards (G), Forwards (F), and Centers (C).

Merged Trends

Goal: Create a cleaned, season-level NBA shot dataset using seasons from 2000-01 to 2023-24 in order to analyze historical shot selection trends.

1. We standardized player names across data sources by converting them to ASCII, and then using str.replace to get rid of unnecessary characters as well as suffixes in player names.
2. We manually mapped eleven names that had not originally been matched but were determined to be matches by our visual inspection (primarily nicknames).
3. The shot dataset was then left-joined with the player-position dataset using season and standardized player name as keys to add the player's position to the shot attempt data.
4. To evaluate the merge we calculated missing position values, finding that only 0.18% of rows failed to merge, consisting of 11 players, which we dropped from our final resulting dataframe.

Data Analysis Challenges & Descriptive Statistics

Challenges

One challenge we faced while merging two datasets was that some players played for more than one team in the same season. In our Basketball Reference dataset, these players were labeled as **2TM**, **3TM**, **4TM**, and **5TM**, indicating multi-team seasons.

We found 1,451 such records out of 11,679 total player-season entries (about 12.4%), which would have created duplicate player-season rows and distorted team-level merges. To prevent double-counting, we merged the datasets using only player and season keys and relied on team information from the shot dataset.

This ensured one clean record per player per season. However, if we could clean and join team information as well, it could have given even more precise results.

count	
TEAM_BBR	
2TM	1356
3TM	92
4TM	2
5TM	1

Descriptive Statistics

Shooting Percentage & Accuracy by Season x Shot zone

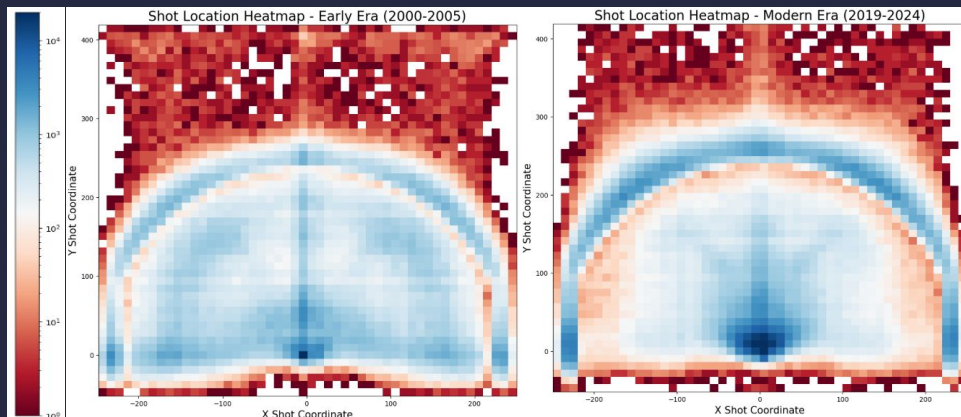
	SEASON	SHOT_ZONES	FREQ	PERCENT	ACCURACY
0	2000-01	Mid-Range	72414	0.380020	0.390767
1	2000-01	Paint	86101	0.451848	0.519971
2	2000-01	Three	32038	0.168132	0.356233
3	2001-02	Mid-Range	71408	0.372000	0.391749
4	2001-02	Paint	86098	0.448528	0.526389
...	...	...	...	...	...
67	2022-23	Paint	107394	0.495417	0.578282
68	2022-23	Three	83622	0.385755	0.362237
69	2023-24	Mid-Range	24588	0.112677	0.418619
70	2023-24	Paint	107713	0.493605	0.574128
71	2023-24	Three	85916	0.393718	0.367452

Shooting Percentage & Accuracy by Position Group x Shot zone

	POS_GROUP	SHOT_ZONES	FREQ	PERCENT	ACCURACY
0	C	Mid-Range	171357	0.223691	0.393360
1	C	Paint	535023	0.698425	0.576452
2	C	Three	59662	0.077883	0.342463
3	F	Mid-Range	501684	0.266324	0.393756
4	F	Paint	880921	0.467646	0.552366
5	F	Three	501129	0.266030	0.357940
6	G	Mid-Range	577134	0.269659	0.401370
7	G	Paint	827933	0.386842	0.519409
8	G	Three	735166	0.343498	0.363204

Analysis and Visualization: Shot Location Heatmaps

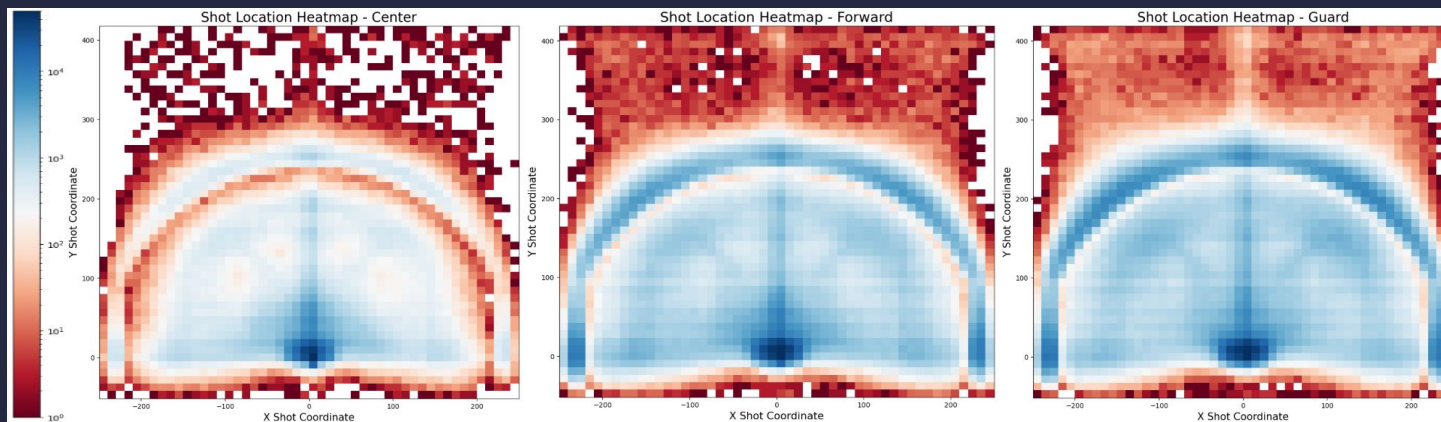
Shot Location Heatmap By Early Era (2000-2005) vs Modern Era (2019-2024)



Shot Location Heatmap: This visualization uses the x and y shot coordinates (LOC_X and LOC_Y) to see how shot locations differ by player position group and Early era (2000–2005) vs Modern era (2019–2024).

Early Era (2000-2005) vs Modern Era (2019-2024): The early era heatmap shows a balanced concentration of shots from three-point area, mid-range area, and near the basket. In the modern era heatmap, the heatmap pattern changes a lot. The strongest concentration is clearly along the three-point area. At the same time, the paint shows a strong concentration. However, the concentration has weakened for mid range area. So, it appears that three-point frequency has increased significantly, paint stayed in similar level, and mid-range shots have decreased significantly.

Shot Location Heatmap By Position Group



By Position Group: Guards show the strongest concentration in the three-point area. The dark blue arc is clearly shown on the heatmap. Forwards show a similar pattern to the Guards. However, it shows more balance across three-point, mid-range, and paint areas. The Centers' heatmap clearly shows a focus in the paint area. It also shows less mid-range and three point emphasis than the other position groups.

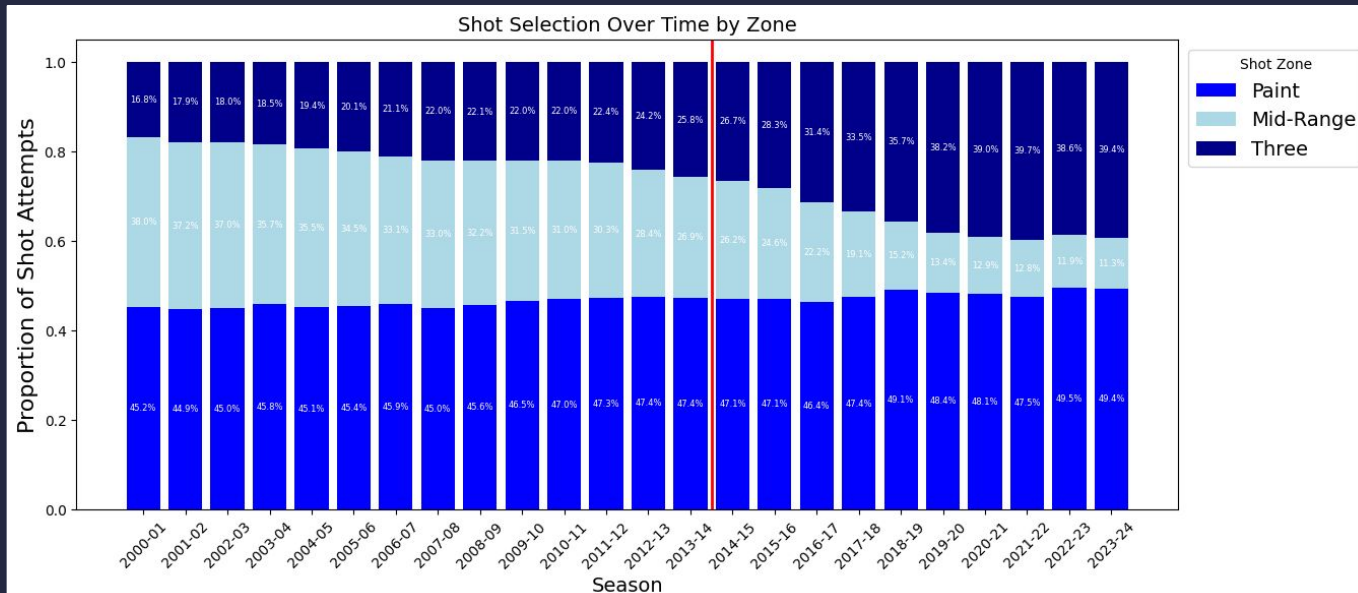
Analysis and Visualization: Shot Selection Over Time

Shot Selection Analysis by Position: A key goal of our analysis was to understand how these shot selection trends may have changed over time and if this differs across guards, forwards, and centers. We first wanted to examine how, if at all, proportions of shots taken in each of our three defined zones changed as the years went on, and then look to see which positions were making these changes. The stacked bar chart below shows the shot selection (based on shots attempted) across all positions from 2000-01 to 2023-24. This allows us to see the big-picture trend where overall proportion of shots taken in the zone “Three” increase and is accompanied by a decrease in “Mid-Range” and a marginal increase in “Paint”.

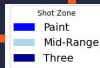
Three point attempts: Looking at the first five years in our analysis (2000-01 through 2004-05) we observe the proportion of shot attempts in the “Three” zone to be under 20%. In the most recent five seasons in our analysis (2019-20 through 2023-24) we observe this proportion to be over 38%, nearly double that of the first five seasons. We mark the change where three-point attempts surpass mid-range by proportion by a red vertical line, between the 2013-14 and 2014-15 seasons.

Mid-Range attempts: The “Mid-range” is the only zone to see a consistent decrease in proportion of shot attempts across our seasons. The first half of our seasons we observe the proportion to hover around a third of shots, although that proportion decreases over the seasons. In the second half of the seasons, this decrease accelerates, and in the final five seasons in this analysis the proportion of attempts in this zone falls to below 14%.

Paint attempts: Shots in the paint start and end as the largest proportion of shots attempted in our seasons of analysis. While this is true, we do observe the earlier seasons to have the proportion closer to 45% of shot attempts while in the later seasons this proportion approaches 50%.



Analysis and Visualization: Shot Selection Over Time By Zone

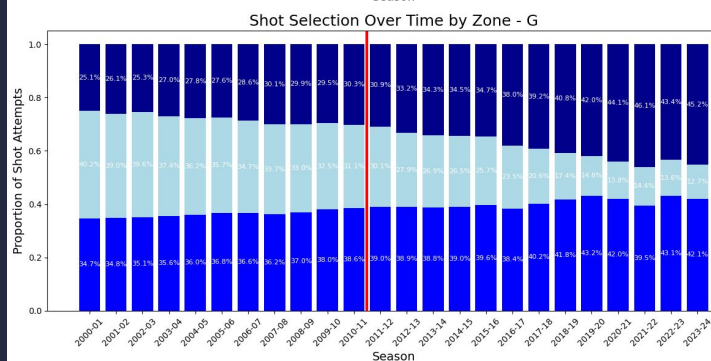
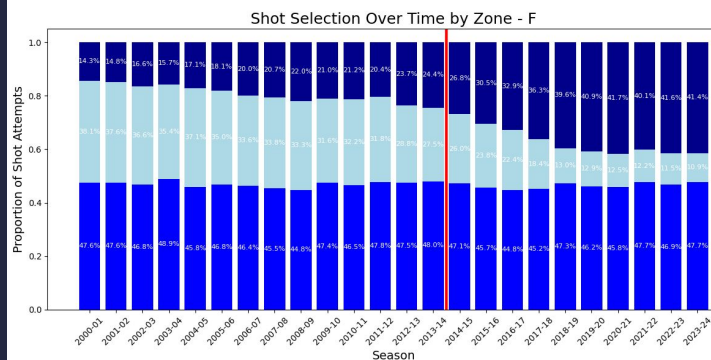
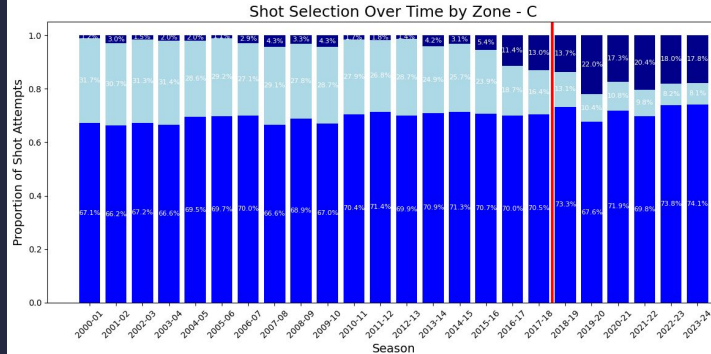


The three charts on the right display the shot selection over time by position showing center (C) at the top, guards (G) in the middle, and forwards (F) at the bottom.

Centers: Over our time frame the proportion of shots taken in the paint by centers has largely remained the same, making up the majority, **between 66-74.1%**. While slight variations in paint and mid-range proportion values are noted in the first 15 seasons, in the 2015-16 season centers reach their first season with **over 5%** of shots taken in the “Three” zone accompanied by the centers’ lowest yet recorded proportion of “Mid-range” shot attempts. This is followed by an increase to **over 10%** of shots attempted by centers in the “Three” zone for the 2016-17 through 2023-24 seasons, even peaking at **over 22%** in one season. The 2018-19 season marks the first season in which three-point attempts make a larger proportion than mid-range for centers, which we visualize with a vertical red line. While the basketball focus of centers is scoring near the basket, the analysis shows that the overall change in shot selection to favor threes is reflected in the shot selection of centers.

Forwards: The first six seasons in our analysis show forward “Mid-range” shots making up **over 35%** of attempts, and the last six seasons show forward “Mid-range” shots making up **less than 13%** of attempts. Forwards also saw a large increase in “Three” zone attempts, and although analysis shows forwards did not shoot **over 20%** of attempts in the “Three” zone until the 2006-07 season, forwards were shooting **over 40%** of their attempts in the “Three” zone for the most recent 5 seasons in this analysis. Forwards see the proportion of “Three” point attempts overtake the “Mid-range” for the first time in the 2014-15 season, which we mark with a vertical red line.

Guards: Over our time frame, analysis shows that guards “Mid-range” zone proportion of shot attempts decrease every single season except for two of them, and while the first six seasons in our analysis show guard “Mid-range” shots making up **over 35%** of attempts, the last five seasons in our analysis show guard “Mid-range” shots making up **less than 18%** of attempts. We mark the first season where the proportion of “Three” zone attempts surpasses “Mid-range” with a vertical red line. This is accompanied by an increase in “Three” zone proportion which begins in the **mid 20s** and ends in the **mid 40s**.



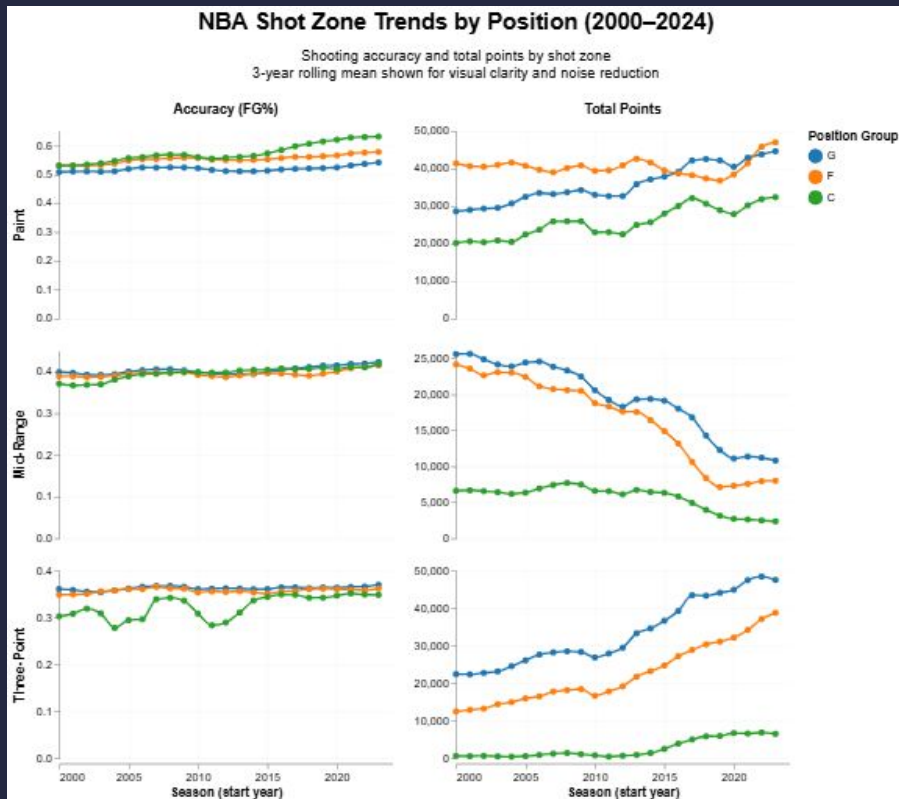
Analysis and Visualization: NBA Shot Zone Trends by Position

NBA Shot Zone Trends by Position

Accuracy: In the paint, mid-range, and three-point zones the accuracy appears to increase over seasons. The paint is the most accurate zone for all positions at both the beginning of our timeframe and the end. The accuracy for the three-point zone does not appear to increase across the timeframe in a manner that is visually significant as we inspect the graph to the right.

Total Points: Mid-range total points decrease across all positions in our time frame. Both paint and three-point zones see an increase across positions in total points, with total points in the three-point zone increasing more dramatically. It is also notable that while centers total points in the paint follows a similar upwards trend as other positions do in the paint, the centers trend sees less significant shifts in both the mid-range and three-points zone.

Takeaways: Total points made from the three-point zones increase across positions but we do not see a corresponding increase in efficiency, suggesting that the total points increase comes as a result of shot zone selection.



Hypothesis Testing

How has the distribution of shots taken in the paint, mid-range, and three-point areas changed from the early 2000s to the present?

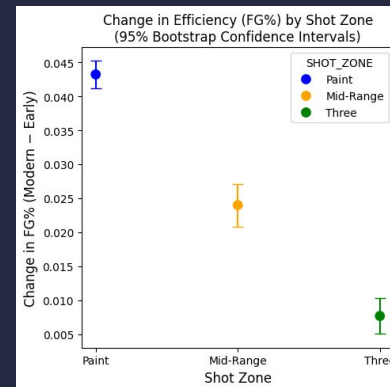
The chi-square test of independence indicates that the distribution of shot locations differs significantly between the early 2000s and the modern era ($\chi^2 = 197,198.6$, $p < .001$). The large chi-squared value shows that the observed (counts in shot zones) deviates far from the expected (the counts we would expect to see if era had no impact on shot zone selection). The effect size, measured by Cramér's V (0.315), also indicates a large and substantively meaningful shift in shot selection.

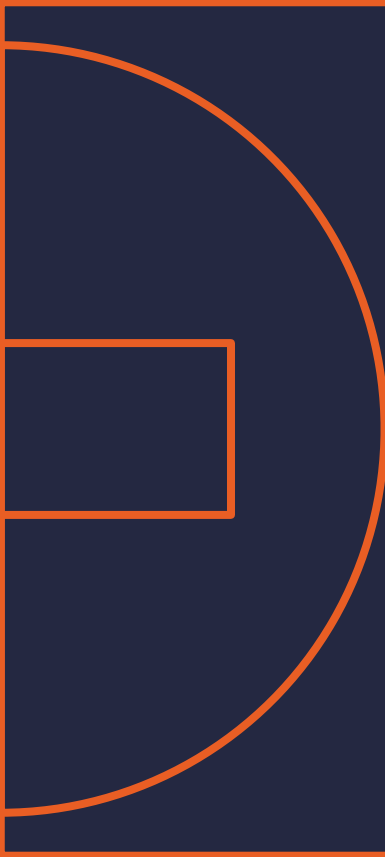
How do these shot selection trends differ across guards, forwards, and centers, and how have positional roles evolved over time?

Position-specific chi-square tests of independence indicate that the distribution of shot locations differs significantly between the early 2000s and the modern era for guards, forwards, and centers (all χ^2 tests, $p < .001$). Effect sizes, measured using Cramér's V , were large across all three position groups (guards: $V = 0.291$; forwards: $V = 0.348$; centers: $V = 0.353$), indicating substantively meaningful shifts in shot selection over time. These results suggest that the evolution in shot selection was not specific to a single position group but instead reflected a league-wide transformation affecting all positions. Across guards, forwards, and centers, mid-range attempts declined sharply, while three-point attempts increased substantially, and paint attempts remained comparatively stable. These effects were particularly dramatic for forwards and centers, highlighting a significant evolution in positional roles and offensive responsibilities over the past two decades.

Are changes in shot selection accompanied by corresponding changes in shot efficiency across different zones?

Bootstrap tests comparing early (2000–2005) and modern (2019–2024) eras indicate that shot efficiency increased across all three shot zones, as measured by field goal percentage (FG%). The largest efficiency gains occurred in the paint, where FG% increased by over four percentage points, suggesting substantial improvements in interior scoring efficiency. Mid-range shots also became more efficient, with moderate increases in FG%, despite a pronounced decline in their overall usage, consistent with improved shot selection. Three-point efficiency increased more modestly on a per-shot basis, with smaller gains in FG%. The plot to the right visualizes these findings with a 95% confidence interval. None of the three intervals contain zero, demonstrating how there is strong evidence in a non-zero change in efficiency for each shot zone. These findings indicate that the modern increase in three-point scoring is driven primarily by changes in shot selection rather than dramatic improvements in shooting accuracy.





Conclusions

Our results show consistent changes across eras in which shot selection has shifted away from mid-range attempts toward three-point shot attempts and shots in the paint. While field goal percentage increased across all shot zones, our bootstrap confidence intervals indicate that these gains, while non-zero, are modest in size including three-point shooting.

Driving Changes in Three Point Scoring: Our bootstrap difference in means testing allowed us to examine efficiency change between the early and modern eras, revealing a non-zero improvement in efficiency across all zones. While each zone improved in efficiency, we found the increase in efficiency was the smallest in the three point zone leading us to conclude that the modern increase in three-point scoring is driven by changes in shot selection rather than large improvements in shooting accuracy.

Our position-based analysis also reinforced these findings. All three positions have seen a proportional increase in shot attempts in the three-point zone. Centers saw a smaller magnitude of change in their three-point and mid-range total points scored than their forward and guard counterparts, but their overall trends over our time frame still reflected those of the other positions, with an increase in total points made in the three-point zone over time and a decrease in total points scored in the mid-range zone over time. All three positions saw the three-point zone overtake the proportion of shots in the mid-range zone. This occurred first for guards, then forwards, and finally centers. This change across all positions demonstrates that while each position may take on a role within the game, shot selection strategy has impacted the shot choices of all positions.

Statement of Work & Sources

Keegan Schmit

Initial dataset upload and dataframe creation. Consolidation of shot zones into our 3 defined zones. Coded the NBA shot Zone Trends by Position visualization and coded and analyzed the hypothesis testing portion of this project. Helped with code cleaning, written analysis of visualizations and hypothesis testing, background, and conclusion.

Brady Hong

Dataset cleaning and manipulation, visually analyzed dataset merge and identified merging based errors. Identified rows that didn't merge in order to analyze the quality of the merge. Completed the descriptive statistics. Coded the heat map visualizations and analyzed results of these visualizations. Helped with code cleaning, written analysis of visualizations, background, and conclusion.

Samantha Hastie

Completed the ReadMe and notebook organization. Analyzed similar studies and reports to our project. Coded the Shot Selection Over Time by Zone visualization and corresponding written analysis. Written analysis of data manipulation/cleaning. Coded visualization of increase in efficiency confidence intervals. Helped with code cleaning, written analysis of visualizations, background, and conclusion.

Collaboration

Our team met over zoom several times to (1) brainstorm ideas and decide a project idea, (2) create a project proposal and delegate project tasks, and (3) review progress and plan next steps. The team stayed active between meetings in our slack channel in which we shared individual updates on what we had completed, collectively answered questions that came up, and discussed new ideas and insights that came up during our analysis. To improve on collaboration in the future, our team could improve on the timeliness of updates when working in the code. This could be beneficial because while working in a shared notebook it is helpful to know exactly when others are working and making changes. Although our updates to teammates were regular and thorough, we could improve collaboration by sharing exactly when we are completing certain sections in the code.

Sources

Freitas, L. "Shot distribution in the NBA: did we see when 3-point shots became popular?". *Ger J Exerc Sport Res* 51, 237–240 (2021).

<https://doi.org/10.1007/s12662-020-00690-7>

Kilcoyne , S. (2020). *The Decline of the Mid-Range Jump Shot in Basketball: A Study of the Impact of Data Analytics on Shooting Habits in the NBA*.

Zajac, Tomasz, et al. "Long-Term Trends in Shooting Performance in the NBA: An Analysis of Two- and Three-Point Shooting across 40 Consecutive Seasons."

International Journal of Environmental Research and Public Health, vol. 20, no. 3, 2023, p. 1924. MDPI, <https://doi.org/10.3390/ijerph20031924>.